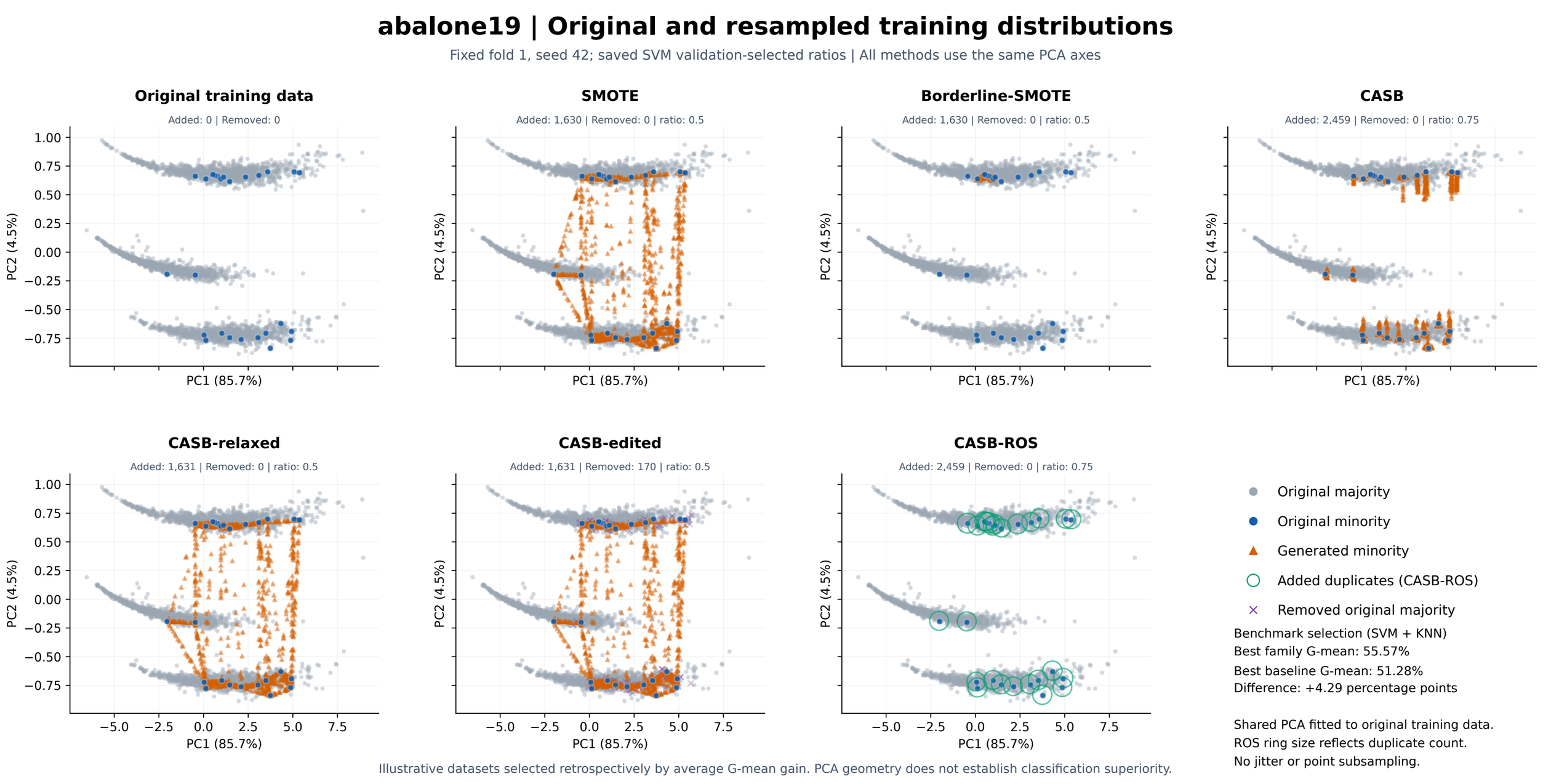

abalone19 | Original and resampled training distributions
Fixed fold 1, seed 42; saved SVM validation-selected ratios | All methods use the same PCA axes
Original training data
Added: 0 | Removed: 0
SMOTE
Added: 1,630 | Removed: 0 | ratio: 0.5
Borderline-SMOTE
Added: 1,630 | Removed: 0 | ratio: 0.5
CASB
Added: 2,459 | Removed: 0 | ratio: 0.75
CASB-relaxed
Added: 1,631 | Removed: 0 | ratio: 0.5
CASB-edited
Added: 1,631 | Removed: 170 | ratio: 0.5
CASB-ROS
Added: 2,459 | Removed: 0 | ratio: 0.75
PC1 (85.7%)
PC2 (4.5%)
Original majority
Original minority
Generated minority
Added duplicates (CASB-ROS)
Removed original majority
Benchmark selection (SVM + KNN)
Best family G-mean: 55.57%
Best baseline G-mean: 51.28%
Difference: +4.29 percentage points
Shared PCA fitted to original training data.
ROS ring size reflects duplicate count.
No jitter or point subsampling.
Illustrative datasets selected retrospectively by average G-mean gain. PCA geometry does not establish classification superiority.